

Math Reference

1 Inline Math

Use dollar signs for inline math: $a^2 + b^2 = c^2$. Subscripts and superscripts: x_i , x^2 , x_i^2 .
Common constants: π , e , ∞ .

2 Display Math

Use `equation` for numbered display math:

$$E = mc^2 \tag{1}$$

Use `equation*` (or `$$...$$`) to suppress numbering:

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

3 Fractions and Roots

$$\frac{a}{b} \quad \frac{x^2 + 1}{2x - 3} \quad \sqrt{x} \quad \sqrt[3]{x}$$

4 Sums, Products, Integrals

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad \prod_{i=1}^n i = n! \quad \int_a^b f(x) dx$$

5 Limits

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \quad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

6 Common Symbols

Relational: $\leq$, $\geq$, $\neq$, $\approx$, $\equiv$, $\propto$

Set notation: $\in$, $\notin$, $\subset$, $\subseteq$, $\cup$, $\cap$, $\emptyset$

Arithmetic: $\pm$, $\mp$, $\times$, $\div$, $\cdot$

Logic: $\forall$, $\exists$, $\neg$, $\wedge$, $\vee$, $\Rightarrow$, $\Leftrightarrow$

7 Aligned Equations

Use `align*` to align multiple equations at `&`:

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x + 1)^2 \end{aligned}$$

8 Matrices

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$